BRIDGING THE TEMPORAL GAP

A Proactive Multi-Agent System for Dynamic Travel Itinerary Re-Planning

Master's Thesis Proposal in Computer Science

THE LLM KNOWLEDGE CUTOFF PROBLEM

❄️ Static Intelligence

Modern Large Language Models (LLMs) are frozen in time post-training. Their "brain" is a snapshot of the world that cannot naturally perceive events happening **today**.

⚡ Real-World Entropy

Urban environments are volatile. Strikes, protests, and accidents emerge in minutes. This creates a "Temporal Gap" where AI advice becomes obsolete or dangerous.

STATIC MAPS VS DYNAMIC REALITY

- **Information Lag:** Standard assistants rely on cached database entries from hours or days ago.
- **The "Last Mile" Blindness:** A museum is "open" according to the AI, but closed due to a sudden protest outside.
- **Passive Response:** Current systems wait for users to discover problems rather than predicting them.

THE THREE-AGENT ARCHITECTURE

Planner

Listens to user desires and synthesizes an initial itinerary based on device metadata and core intent.

Observer

Asynchronously monitors live API streams, news, and social signals for disruptive real-time events.

Mind

The reasoning orchestrator that evaluates if observations require a plan modification (The Trigger).

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AGENT 1: THE PLANNER

- **Intent Decoding:** Extracts semantic goals from natural language (e.g., "I want to visit the center of Rome and I would like to eat near the Trevi Fountain").
- **Temporal Scheduling:** Maps activities to a flexible time-graph, considering travel time and typical opening hours.
- **Initial Grounding:** Validates the baseline feasibility of locations using verified geolocalization data.

AGENT 2: THE OBSERVER

- **Heterogeneous Streams:** Monitors News APIs, Transit Alerts, and Social Media (X/Twitter) for high-frequency updates.
- **Semantic Filtering:** Uses LLM-based classification to distinguish between "noise" and "disruptive urban events."
- **Continuous Polling:** Operates as an asynchronous background task to ensure zero-latency detection of emergencies.

AGENT 3: THE MIND

- **Impact Assessment:** Calculates the geometric and semantic intersection between a "Threat" and the "Path"
- **Conflict Resolution:** Decides priority when data sources conflict (e.g., Official API says 'Open' but News says 'Blocked').
- **Proactive Dispatch:** Triggers notifications to the user before they encounter the obstacle.

THE TECHNICAL STACK

Frameworks

- Python
- FastAPI (Backend Integration)
- LLM (Reasoning)